When a Normal Curve Distorts the Tail

AP Statistics · Unit 3 · Topic 3.3 · B: 2026-11-24 · E: 2026-12-23 · TEACHER KEY

CED TETHER

- **VAR-1.H** Identify questions suggested by variation in the shapes of distributions of samples taken from the same population. [**Skill 1.A**]
- **VAR-1.H.1** Variation in shapes of data distributions may be random or not.

Essential question: When should a simulated sampling distribution be trusted more than a normal approximation?

DOK-3 skill: 4.B · **FRQ pattern:** `normal-approximation-vs-simulation-for-a-skewed-tail`

DOK rationale: Part (c) coordinates an expected-count check, the simulated distribution's shape, and two tail estimates to choose a defensible method, bound the evidence, and explain the consequence of the rejected analysis.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Let students choose between the two reported tail probabilities without confirming a method.
- Begin the combined 6.1–6.2 video follow-along at minute 5.
- Return to the board slide at minute 33. Circulate and require a shape or expected-count check before a probability choice.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose the normal-model or simulation estimate and cite one visible feature.
- Complete the follow-along about chance explanations, normal models, and intervals for a proportion.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- What does one bar in this simulated distribution count?
- Which expected count warns you that a symmetric curve may miss the tail?
- Why do simulated samples with more than six activations still count?
- Can the two methods disagree in size yet lead to the same 5 percent decision?

Adult role

Solo teacher. Circulate 5 pairs. Ask for the model, shape check, tail event, and reader consequence; do not accept a method choice based only on which probability is smaller.

[WARN] WATCH FOR

- Treating every sampling distribution as normal regardless of expected counts.
- Counting exactly six activations instead of six or more.
- Saying a small chance probability proves why the activation rate differs.

SCORING PART (c) — E / P / I

Expected elements

- **selects-matched-simulation** (required) — Selects Jules's simulation estimate rather than Mara's unsupported normal approximation
- **uses-shape-or-count-evidence** (required) — Cites the right-skewed discrete shape or the failed expected counts 2.4 and 9.6
- **uses-tail-and-bounds-conclusion** (required) — Uses 4 of 200, or 0.020, to make the 5 percent evidence decision without claiming proof of a cause
- **states-reader-consequence** (required) — Explains that 0.005 makes the result appear about four times rarer and overstates the evidence

E	Chooses the matched simulation, supports that choice with the failed normal condition or displayed shape, uses 0.020 for the 5 percent decision, and explains how 0.005 overstates rarity without claiming causation.
P	Chooses the simulation and reaches the correct evidence decision but omits either the shape or expected-count support, the quantitative comparison, or the consequence for a reader.
I	Chooses the normal result solely because it is smaller, treats the distribution as automatically normal, counts only exactly six incorrectly, or states that the result proves a cause.

Common mistakes

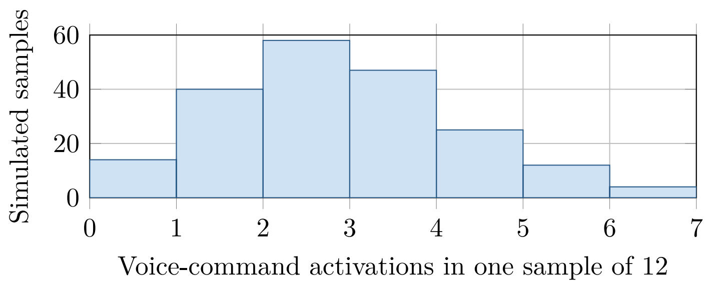
- Using $6/12 = 0.50$ as the chance probability
- Counting only the bar at 6 without recognizing that all larger results would also be at least as extreme
- Checking observed successes and failures instead of expected counts under p_0 for this model
- Calling the two percent estimate proof that the developer intentionally misstated the rate

[STAR] TODAY'S PROBLEM — annotated

Suppose an app developer's model says that $p_0 = 0.20$ of all new users activate voice commands during their first day. A simple random sample of 12 new users includes 6 who activate the feature, so $\hat{p} = 0.50$. Under the developer's model, a class simulates 200 samples of 12 independent users; the histogram shows the number of activations in each sample.

Mara: "A normal model for $\hat{p}$ gives $P(\hat{p} \geq 0.50) \approx 0.005$, so that is the chance probability I would report."

Jules: "The simulated distribution is not very normal. I would estimate the chance probability directly from its tail."



FIRST TAKE (5 min)

Which chance probability would you report, Mara's or Jules's? Cite one detail from the histogram or sample size.
Key: Not graded. Preserve the normal-model versus matched-simulation tension.

Rules callout "CHECK THE SHAPE BEFORE USING NORMAL (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** State the most common number of activations in the simulated samples and describe the distribution as roughly symmetric, skewed left, or skewed right.

Key: Two activations is most common, occurring in 58 of the 200 simulations. The distribution is discrete and skewed right, with most results from 0 through 4 and a thinning tail toward 6.

DOK 2 **(b)** Use the histogram to estimate $P(\hat{p} \geq 0.50)$ under $p_0 = 0.20$. Then calculate np_0 and $n(1-p_0)$ and assess the expected-count condition for a normal approximation.

Key: Six activations out of 12 gives $\hat{p} = 0.50$. Four simulations have 6 or more activations, so the simulated estimate is $4/200 = 0.020$. The expected counts are $np_0 = 12(0.20) = 2.4$ and $n(1 - p_0) = 12(0.80) = 9.6$; neither reaches 10, so the normal-approximation condition is not met.

DOK 3 **(c)** Choose the better chance-probability method and use a 5% criterion to judge evidence against $p_0 = 0.20$. Cite the simulated tail and a shape or expected-count feature, compare 0.020 with 0.005, and explain the reader consequence of Mara's report.

Key: Use Jules's simulation: expected counts 2.4 and 9.6 fail the normal check, and the distribution is discrete and right-skewed. Its tail is $4/200 = 0.020 < 0.05$, convincing evidence against $p_0 = 0.20$ in the higher direction if the random sample is sound, but not proof of a cause. Mara's 0.005 is one fourth as large, making the result appear four times rarer and overstating the evidence. The exact binomial tail, about 0.0194, supports the simulation.

EXIT

One thing the video changed about my first take: ____________________